

0620 • PAPER 4

TOPIC

03

Stoichiometry

Questions only • Original examination pages retained

33

QUESTIONS

467

TOTAL MARKS

1

SECTIONS



USE THE TRACKER • ATTEMPT WITHOUT NOTES • MARK • REVIEW ERRORS

TOPIC CONTENTS

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01

Moles, formulae, equations and chemical calculations

33 questions • 467 marks

3.1 Moles, formulae, equations and chemical calculations

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3.1 Moles, formulae, equations and chemical calculations

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3.1 Moles, formulae, equations and chemical calculations

#	SOURCE	MARKS	DONE	MARK	REVIEW
29	2025 May/June Paper 43 Question 4	13	☐	☐	☐
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32	2025 Oct/Nov Paper 42 Question 3	13	☐	☐	☐
33	2025 Oct/Nov Paper 43 Question 4	17	☐	☐	☐

4 This question is about reactions of bases and acids.

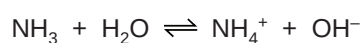
(a) Ammonia is a gas at room temperature.

What is the test for ammonia gas? Describe the positive result of this test.

test

result [2]

(b) Ammonia reacts with water to form ions.



(i) How does this equation show that ammonia, NH_3 , behaves as a base?

..... [1]

(ii) Aqueous ammonia is described as a weak base.

Suggest the pH of aqueous ammonia.

pH = [1]

(iii) Describe what is seen when aqueous ammonia is added to aqueous copper(II) sulfate, until no further change is seen.

.....

.....

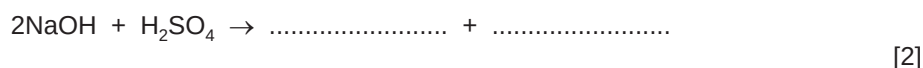
..... [3]

(c) Aqueous sodium hydroxide, NaOH(aq), is a strong alkali that reacts with dilute sulfuric acid exothermically.

(i) What type of reaction is this?

..... [1]

(ii) Complete the equation for the reaction between aqueous sodium hydroxide and dilute sulfuric acid.



(d) A student wanted to find the concentration of some dilute sulfuric acid by titration. The student found that 25.0 cm³ of 0.0400 mol/dm³ NaOH(aq) reacted exactly with 20.0 cm³ of H₂SO₄(aq).

(i) Name a suitable indicator to use in this titration.

..... [1]

(ii) Calculate the concentration of the H₂SO₄(aq) in mol/dm³ using the following steps.

- Calculate the number of moles of NaOH in 25.0 cm³.

moles =

- Deduce the number of moles of H₂SO₄ that reacted with the 25.0 cm³ of NaOH(aq).

moles =

- Calculate the concentration of H₂SO₄(aq) in mol/dm³.

concentration = mol/dm³ [3]

(iii) Calculate the concentration of the 0.0400 mol/dm³ NaOH(aq) in g/dm³.

concentration = g/dm³ [2]

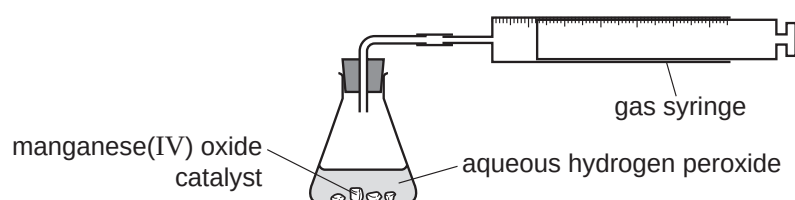
[Total: 16]

- 4 Oxygen is produced by the decomposition of hydrogen peroxide. Manganese(IV) oxide is the catalyst for this reaction.

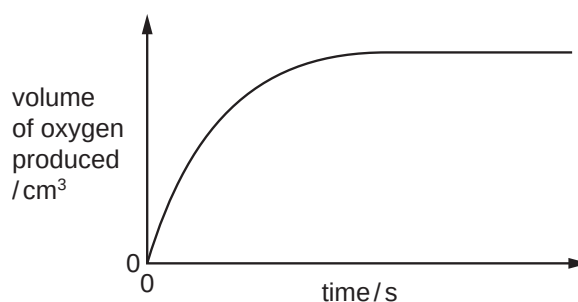
(a) What is meant by the term *catalyst*?

.....
.....
..... [2]

- (b) A student measures the volume of oxygen produced at regular time intervals using the apparatus shown. Large lumps of manganese(IV) oxide are used.



A graph of the results is shown.



What happens to the **rate** of this reaction as time increases?
In your answer, explain why the rate changes in this way.

.....
.....
.....
.....
..... [4]

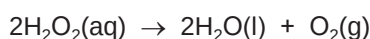
- (c) The experiment is repeated using the same mass of manganese(IV) oxide. Powdered manganese(IV) oxide is used instead of large lumps. All other conditions stay the same.

Sketch a graph on the axes in (b) to show how the volume of oxygen changes with time. [2]

- (d) In terms of particles, explain what happens to the rate of this reaction when the temperature is increased.

.....
.....
.....
.....
..... [3]

- (e) The equation for the decomposition of hydrogen peroxide is shown.



25.0 cm³ of aqueous hydrogen peroxide forms 48.0 cm³ of oxygen at room temperature and pressure (r.t.p.).

Calculate the concentration of aqueous hydrogen peroxide at the start of the experiment using the following steps.

- Calculate the number of moles of oxygen formed.

..... mol

- Deduce the number of moles of hydrogen peroxide that decomposed.

..... mol

- Calculate the concentration of hydrogen peroxide in mol/dm³.

..... mol/dm³
[3]

- (f) Oxygen can also be produced by the decomposition of potassium chlorate(V), KClO₃.

The only products of this decomposition are potassium chloride and oxygen.

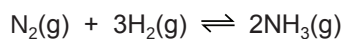
Write a chemical equation for this decomposition.

..... [2]

[Total: 16]

2 Ammonia is manufactured by the Haber process.

(a) The equation for the reaction is shown.



(i) State what is meant by the symbol $\rightleftharpoons$.

..... [1]

(ii) State **one** source of hydrogen used in the manufacture of ammonia.

..... [1]

(b) The table shows some data for the production of ammonia.

pressure / atm	temperature / °C	percentage yield of ammonia
250	350	58
100	450	28
400	450	42
250	550	20

Deduce the effect on the percentage yield of ammonia of:

- increasing the pressure of the reaction

.....

- increasing the temperature of the reaction.

.....

[2]

(c) Explain, in terms of particles, what happens to the rate of this reaction when the temperature is increased.

.....

.....

.....

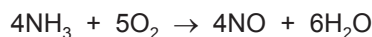
.....

.....

..... [3]

(d) Ammonia, NH_3 , is used to produce nitric acid, HNO_3 . This happens in a three-stage process.

Stage 1 is a redox reaction.



(i) Identify what is oxidised in **stage 1**.

Give a reason for your answer.

substance oxidised

reason

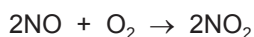
[2]

(ii) In this reaction the predicted yield of NO is 512 g. The actual yield is 384 g.

Calculate the percentage yield of NO in this reaction.

percentage yield of NO = [1]

(iii) The equation for the reaction in **stage 2** is shown.

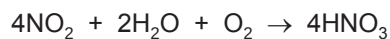


Which major environmental problem does NO_2 cause if it is released into the atmosphere?

.....

..... [1]

(iv) The equation for the reaction in **stage 3** is shown.



Calculate the volume of O_2 gas, at room temperature and pressure (r.t.p.), needed to produce 1260 g of HNO_3 .
Use the following steps.

- Calculate the number of moles of HNO_3 .

moles of HNO_3 =

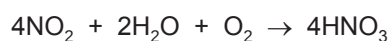
- Deduce the number of moles of O_2 that reacted.

moles of O_2 =

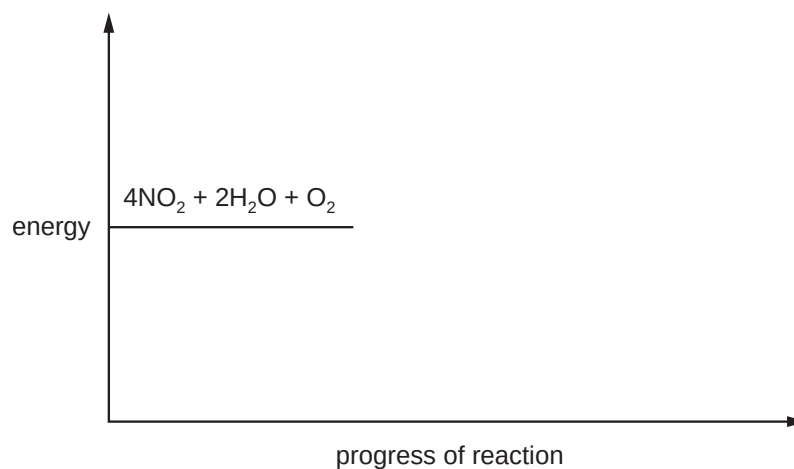
- Calculate the volume of O_2 gas that reacts at room temperature and pressure (r.t.p.).

volume of O_2 gas = dm^3
[4]

(e) The reaction in **stage 3** is exothermic.



Complete the energy level diagram for this reaction. Include an arrow that clearly shows the energy change during the reaction.

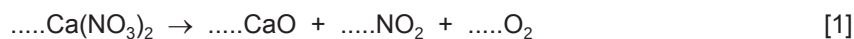


[3]

[Total: 18]

6 (a) An endothermic reaction occurs when calcium nitrate is heated.

(i) Balance the equation for this reaction.



(ii) State the type of reaction shown by the equation.

..... [1]

(b) Describe the test for a nitrate ion.

test

.....

result

.....

[3]

[Total: 5]

6 (a) Ethane, propane and butane are members of the same homologous series.

(i) Name this homologous series.

..... [1]

(ii) State **two** ways members of the same homologous series are similar.

1

2

[2]

(b) One mole of ethane, C_2H_6 , contains 6.02×10^{23} molecules.

Calculate how many molecules are in 15 g of ethane.

number of ethane molecules = [1]

(c) Propane reacts with chlorine.

(i) Write the formula of the product which does not contain carbon.

..... [1]

(ii) Draw the structure of an organic product formed. Show all of the atoms and all of the bonds.

[1]

(iii) State the name of this type of reaction.

..... [1]

- (d) (i) Aqueous bromine was added to a sample of ethene.

Give the colour change seen.

from to [2]

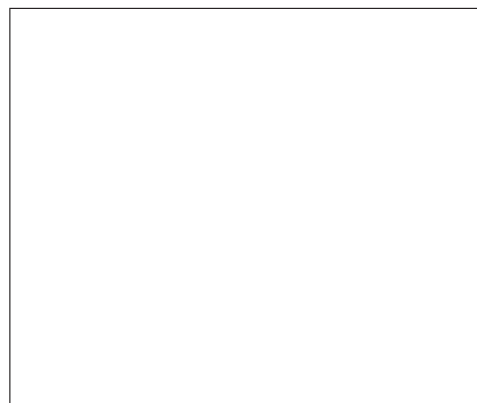
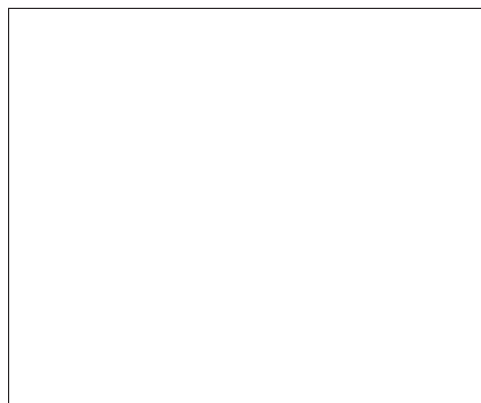
- (ii) Explain, in terms of bonding, why there is no colour change when aqueous bromine is added to ethane.

.....

..... [1]

- (e) There are two structural isomers with the formula C_4H_{10} .

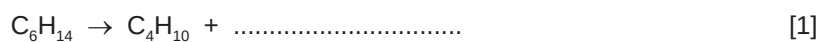
- (i) Draw the structures of both of these isomers, showing all of the atoms and all of the bonds.



[2]

- (ii) Butane is formed when longer chain hydrocarbons are cracked.

Complete the chemical equation to show the other product when butane is formed by cracking.



(f) A compound contains 85.7% carbon and 14.3% hydrogen by mass.

(i) Calculate the empirical formula of this compound.

Show your working.

..... [2]

(ii) The molecular mass of the compound is 112.

Calculate the molecular formula of this compound.

..... [1]

[Total: 16]

- 4 Zinc is manufactured from zinc blende. Zinc blende is an ore which consists mainly of zinc sulfide, ZnS.

- (a) Zinc blende is roasted in air. One of the products is zinc oxide.

Name the **other** product formed in this reaction.

..... [1]

- (b) Zinc oxide is then converted into zinc.

Zinc oxide and coke, a source of carbon, are heated in a furnace. Hot air is blown into the furnace.

- (i) Give **two** reasons why coke is needed.

1

2 [2]

- (ii) Write a chemical equation for the formation of zinc in the furnace.

..... [1]

- (iii) Zinc has a melting point of 420 °C and a boiling point of 907 °C. The temperature inside the furnace is 1200 °C.

Explain how this information shows that the zinc produced inside the furnace is a gas.

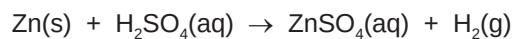
..... [1]

- (iv) The gaseous zinc is converted to molten zinc.

Name this change of state.

..... [1]

- (c) Zinc reacts with dilute sulfuric acid to produce aqueous zinc sulfate.



Hydrated zinc sulfate crystals are made from aqueous zinc sulfate.

Step 1 Solid zinc is added to dilute sulfuric acid until zinc is in excess.

Step 2 Excess zinc is separated from aqueous zinc sulfate by filtration.

Step 3 Aqueous zinc sulfate is heated until the solution is saturated.

Step 4 The saturated solution is allowed to cool and crystallise.

Step 5 The crystals are removed and dried.

- (i) Name the residue in **step 2**.

..... [1]

- (ii) In **step 3**, a saturated solution is produced.

Describe what a saturated solution is.

.....
.....
..... [2]

- (iii) Name **two** compounds each of which react with dilute sulfuric acid to produce aqueous zinc sulfate.

1
2
[2]

(d) When hydrated magnesium sulfate crystals, $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$, are heated they give off water.



A student carries out an experiment to determine the value of x in $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$.

Step 1 Hydrated magnesium sulfate crystals were weighed.

Step 2 Hydrated magnesium sulfate crystals were heated.

Step 3 The remaining solid was weighed.

(i) Describe how the student can ensure that all the water is given off.

.....
.....
..... [2]

(ii) In an experiment, all the water was removed from 1.23 g of $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$. The mass of MgSO_4 remaining was 0.60 g.

M_r : $\text{MgSO}_4 = 120$; M_r : $\text{H}_2\text{O} = 18$

Determine the value of x using the following steps.

- Calculate the number of moles of MgSO_4 remaining.

moles of $\text{MgSO}_4 =$

- Calculate the mass of H_2O given off.

mass of $\text{H}_2\text{O} =$ g

- Calculate the moles of H_2O given off.

moles of $\text{H}_2\text{O} =$

- Determine the value of x .

$x =$
[4]

[Total: 17]

- 2 The elements shown are gases at room temperature and pressure.

hydrogen
nitrogen
oxygen
chlorine

- (a) State which **one** of these gases is green.

..... [1]

- (b) The gases shown exist as diatomic molecules.

State the name of **another** element which has diatomic molecules and is a gas at room temperature and pressure.

..... [1]

- (c) When separate samples of each of these gases are placed in a container they will diffuse.

- (i) Describe why these gases diffuse.

..... [1]

- (ii) State which of these four gases has the highest rate of diffusion.

Explain your answer.

gas

explanation

..... [2]

- (d) Nitrogen, oxygen and other substances are found in clean, dry air.

- (i) State the percentage of nitrogen in clean, dry air.

..... [1]

- (ii) Other than nitrogen and oxygen, identify another element found in clean, dry air.

..... [1]

- (iii) Identify a compound found in clean, dry air.

..... [1]

- (iv) Nitrogen and oxygen can be separated from liquid air.

State the name of this process.

..... [2]

[Total: 10]

5 This question is about salts.

(a) Salts that are insoluble in water are made by precipitation.

- Lead(II) iodide, PbI_2 , is insoluble in water.
- All nitrates are soluble in water.
- All sodium salts are soluble in water.

You are provided with solid lead(II) nitrate, $\text{Pb}(\text{NO}_3)_2$, and solid sodium iodide, NaI .

Describe how you would make a pure sample of lead(II) iodide by precipitation.

Your answer should include:

- practical details
- a chemical equation for the precipitation reaction.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

(b) Nitrates decompose when heated.

(i) When hydrated zinc nitrate is heated, oxygen gas is given off.

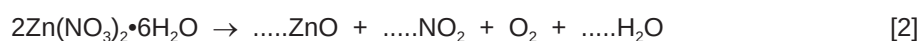
Describe a test for oxygen.

test

observations

[2]

(ii) Complete the equation for the decomposition of hydrated zinc nitrate.



(c) Some sulfates are hydrated.

When hydrated sodium sulfate crystals, $\text{Na}_2\text{SO}_4 \cdot x\text{H}_2\text{O}$, are heated, they give off water.



A student carries out an experiment to determine the value of x in $\text{Na}_2\text{SO}_4 \cdot x\text{H}_2\text{O}$.

step 1 Hydrated sodium sulfate crystals are weighed.

step 2 The hydrated sodium sulfate crystals are then heated.

step 3 The remaining solid is weighed.

(i) Describe how the student can check that all the water has been given off.

.....
.....
..... [2]

- (ii) In an experiment, 1.61 g of $\text{Na}_2\text{SO}_4 \cdot x\text{H}_2\text{O}$ is heated until all the water is given off. The mass of Na_2SO_4 remaining is 0.71 g.

[M_r : Na_2SO_4 , 142; H_2O , 18]

Determine the value of x using the following steps.

- Calculate the number of moles of Na_2SO_4 remaining.

..... mol

- Calculate the mass of H_2O given off.

..... g

- Calculate the number of moles of H_2O given off.

..... mol

- Determine the value of x .

$x =$
[4]

[Total: 15]

7 Many organic compounds contain carbon, hydrogen and oxygen only.

(a) An organic compound **V** has the following composition by mass.

C, 48.65%; H, 8.11%; O, 43.24%

Calculate the empirical formula of compound **V**.

empirical formula = [3]

(b) Compound **W** has the empirical formula CH_4O and a relative molecular mass of 32.

Calculate the molecular formula of compound **W**.

molecular formula = [1]

(c) Compounds **X** and **Y** have the same general formula.

X and **Y** are both carboxylic acids.

Compound **X** has the molecular formula $\text{C}_2\text{H}_4\text{O}_2$.

Compound **Y** has the molecular formula $\text{C}_4\text{H}_8\text{O}_2$.

(i) Deduce the general formula of compounds **X** and **Y**.

..... [1]

- (ii) Draw the structure of compound Y. Show all of the atoms and all of the bonds.

Name compound Y.

name [3]

- (iii) Give the name used to describe a 'family' of similar compounds with the same general formula, similar chemical properties and the same functional group.

..... [1]

- (d) Propene is an unsaturated hydrocarbon. The formula of propene is shown.



- (i) State the colour change observed when propene is added to aqueous bromine.

from to [1]

- (ii) Propene can be produced by cracking long chain alkanes.

Pentadecane, $\text{C}_{15}\text{H}_{32}$, is cracked to produce an alkane and propene in a 1:2 molar ratio.

Complete the chemical equation for this reaction.



- (iii) Propene can be converted into poly(propene).

Name the type of polymerisation that occurs when propene is converted into poly(propene).

..... [1]

- (iv) Complete the diagram to show a section of poly(propene).



[2]

[Total: 15]

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The Periodic Table of Elements

Group																										
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII									
3 Li lithium 7		Key atomic number atomic symbol name relative atomic mass										2 He helium 4														
11 Na sodium 23	12 Mg magnesium 24	4 Be beryllium 9																								
13 Al aluminum 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40																					
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84									
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131									
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —									
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Lv livermorium —	116 Ts tennessine —	117 Og oganeson —	118 Uu unbinilium —									
lanthanoids																	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175					
																	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —
actinoids																	104 Pu plutonium 244	105 Am americium 243	106 Cm curium 247	107 Bk berkelium 247	108 Cf californium 251	109 Es einsteinium 252	110 Fm fermium 257	111 Md mendelevium 258	112 No nobelium 259	113 Lr lawrencium 262

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

3 Sodium hydrogencarbonate is found in baking powder.

When sodium hydrogencarbonate is heated it forms three products.



(a) Name the type of reaction that takes place when sodium hydrogencarbonate reacts in this way.

..... [1]

(b) Calculate the volume of carbon dioxide formed at room temperature and pressure when 12.6 g of NaHCO_3 is heated using the following steps:

- determine the mass of one mole of NaHCO_3

..... g

- calculate the number of moles of NaHCO_3 used

..... moles

- determine the number of moles of carbon dioxide formed

..... moles

- calculate the volume of carbon dioxide formed at room temperature and pressure.

..... dm^3
[4]

(c) Limewater is aqueous calcium hydroxide. Carbon dioxide turns limewater milky because a white precipitate forms.

Write the formula of:

- calcium hydroxide
- the white precipitate that forms when limewater turns milky.
[2]

[Total: 7]

7 Many organic compounds contain carbon, hydrogen and oxygen only.

(a) An organic compound **R** has the following composition by mass.

C, 69.77%; H, 11.63%; O, 18.60%

Calculate the empirical formula of compound **R**.

empirical formula = [2]

(b) Compound **S** has the empirical formula CH_2O and a relative molecular mass of 60.

Calculate the molecular formula of compound **S**.

molecular formula = [2]

(c) Compounds **T** and **V** have the same molecular formula, $\text{C}_3\text{H}_6\text{O}_2$.

- Compound **T** is an ester.
- Compound **V** contains a $-\text{COOH}$ functional group.

(i) State the name given to compounds with the same molecular formula but different structures.

..... [1]

(ii) Name the homologous series that **V** is a member of.

..... [1]

(iii) Draw a structure of compound **T**. Show all of the atoms and all of the bonds.

Name compound **T**.

name [3]

(iv) Draw the structure of compound **V**. Show all of the atoms and all of the bonds.

Name compound **V**.

name [2]

(d) Ethanol can be produced from long chain alkanes such as decane, $C_{10}H_{22}$, in a two-step process.



For each of the two steps:

- name the type of chemical reaction that occurs
- write a chemical equation.

step 1: decane to ethene

type of reaction

chemical equation

step 2: ethene to ethanol

type of reaction

chemical equation [4]

[Total: 15]

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The Periodic Table of Elements

Group												
I	II	Key					III	IV	V	VI	VII	VIII
		1 H hydrogen 1										2 He helium 4
3 Li lithium 7	4 Be beryllium 9	atomic number atomic symbol name relative atomic mass					5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24						13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	50 Sn tin 119	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	82 Pb lead 207	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	114 Fl flerovium —	—

lanthanoids	57	La	lanthanum	139	58	Ce	cerium	140	59	Pr	praseodymium	141	60	Nd	neodymium	144	61	Pm	promethium	—	62	Sm	samarium	150	63	Eu	europium	152	64	Gd	gadolinium	157	65	Tb	terbium	159	66	Dy	dysprosium	163	67	Ho	holmium	165	68	Er	erbium	167	69	Tm	thulium	169	70	Yb	ytterbium	173	71	Lu	lutetium	175
	89	Ac	actinium	227	90	Th	thorium	232	91	Pa	protactinium	231	92	U	uranium	238	93	Np	neptunium	—	94	Pu	plutonium	—	95	Am	americium	—	96	Cm	curium	—	97	Bk	berkelium	—	98	Cf	californium	—	99	Es	einsteinium	—	100	Fm	fermium	—	101	Md	mendelevium	—	102	No	nobelium	—	103	Lr	lawrencium	—

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

2 Acids are important laboratory chemicals.

(a) Some acids completely dissociate in water to form ions.

(i) State the term applied to acids that completely dissociate in water.

..... [1]

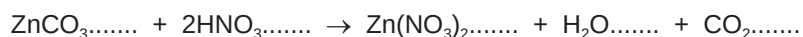
(ii) Complete the equation to show the complete dissociation of sulfuric acid in water.

$\text{H}_2\text{SO}_4 \rightarrow$ [2]

(iii) State the colour of methyl orange in sulfuric acid.

..... [1]

(b) The equation for the reaction between powdered zinc carbonate and dilute nitric acid is shown.



(i) Complete the equation by adding state symbols. [2]

(ii) A student found that 2.5 g of zinc carbonate required 20 cm³ of dilute nitric acid to react completely.

Calculate the concentration of dilute nitric acid using the following steps:

- calculate the mass of 1 mole of ZnCO_3

..... g

- calculate the number of moles of ZnCO_3 reacting

..... moles

- determine the number of moles of HNO_3 reacting

..... moles

- calculate the concentration of HNO_3 .

..... mol/dm³
[4]

[Total: 10]

- 3 Lead is a metallic element in Group IV. One of the ores of lead is galena, which is an impure form of lead(II) sulfide, PbS .

Lead also occurs in the ore cerussite, which contains lead(II) carbonate, PbCO_3 .

- (a) Calculate the relative formula mass, M_r , of PbCO_3 .

M_r of PbCO_3 = [1]

- (b) The M_r of PbS is 239.

Calculate the percentage of lead by mass in PbS .

percentage of lead by mass in PbS = [1]

- (c) The percentage of lead by mass in PbCO_3 is 77.5%.

Use this information and your answer to (b) to suggest whether it would be better to extract lead from PbCO_3 or PbS .

Give a reason for your answer.

.....
..... [1]

- (d) When lead(II) carbonate is heated it decomposes into lead(II) oxide, PbO , and carbon dioxide.

Write a chemical equation for this reaction.

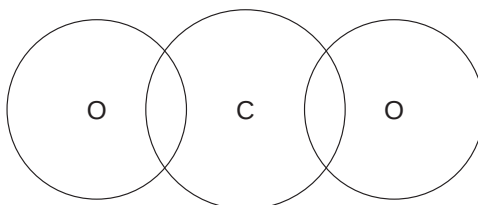
..... [1]

- (e) Lead(II) carbonate reacts with dilute nitric acid. One of the products is aqueous lead(II) nitrate, $\text{Pb}(\text{NO}_3)_2$.

Write a chemical equation for this reaction.

..... [2]

- (f) Lead(II) oxide and carbon dioxide are oxides of Group IV elements.
- (i) Complete the diagram to show the electron arrangement in one molecule of CO_2 .
Show only the outer electrons.



[2]

- (ii) The melting points of lead(II) oxide and carbon dioxide are shown.

	melting point/ $^{\circ}\text{C}$
lead(II) oxide	886
carbon dioxide	-56

Use your knowledge of structure and bonding to explain why lead(II) oxide has a much higher melting point than carbon dioxide.

Your answer should refer to:

- the types of particles involved
- the relative strength of the forces of attraction between the particles.

.....

.....

.....

..... [3]

(g) Part of the reactivity series is shown.

magnesium	most reactive
lead	↑
copper	least reactive

Aqueous lead(II) nitrate contains Pb^{2+} ions.

Two experiments are carried out.

In Experiment 1, magnesium is added to aqueous lead(II) nitrate.

In Experiment 2, copper is added to aqueous lead(II) nitrate.

Write an ionic equation for any reaction that occurs in each experiment. If no reaction occurs write 'no reaction'.

Experiment 1

Experiment 2 [2]

(h) When lead(II) nitrate is heated it decomposes to produce the same gaseous products as when copper(II) nitrate is heated.

(i) One of the gaseous products is oxygen.

Describe a test for oxygen.

test

observations [2]

(ii) Name the other gaseous product.

..... [1]

[Total: 16]

3 Nitrogen dioxide, NO_2 , is an atmospheric pollutant and is formed in car engines.

(a) Explain how nitrogen dioxide is formed in car engines.

.....
..... [2]

(b) Nitrogen dioxide causes respiratory problems.

State one **other** adverse effect of nitrogen dioxide.

..... [1]

(c) Nitrogen dioxide emissions can be reduced by adding an aqueous solution of urea, $(\text{NH}_2)_2\text{CO}$, to car exhaust gases.

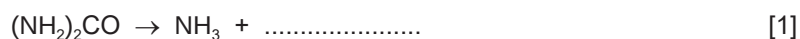
The heat of the exhaust gases breaks down the urea into simpler substances.

(i) Name the type of reaction which occurs when a substance is heated and breaks down into simpler substances.

..... [1]

(ii) One molecule of urea breaks down to form one molecule of ammonia and one other molecule.

Complete the chemical equation to show the formula of the other molecule formed in this reaction.



(iii) State the test for ammonia.

test

observations [2]

(d) The ammonia formed reacts with nitrogen dioxide to form nitrogen and water.

(i) Balance the equation for this reaction.



(ii) State how the equation shows that the nitrogen in nitrogen dioxide is reduced.

..... [1]

(iii) This reaction is a redox reaction.

State the meaning of the term *redox*.

..... [1]

(e) 135 moles of urea, $(\text{NH}_2)_2\text{CO}$, is stored in the tank of a car.

Calculate the mass, in kg, of the stored $(\text{NH}_2)_2\text{CO}$.

mass of $(\text{NH}_2)_2\text{CO}$ = kg
[2]

(f) Another oxide of nitrogen formed in car engines is nitrogen monoxide, NO. A catalytic converter removes NO by reacting it with a gas formed by incomplete combustion of the fuel. Two non-toxic gases are formed.

(i) Name the gas formed by incomplete combustion of the fuel.

..... [1]

(ii) Name the **two** non-toxic gases formed.

..... and [1]

[Total: 15]

3 This question is about sodium and compounds of sodium.

(a) (i) Describe the bonding in a metallic element such as sodium.

You may include a diagram as part of your answer.

.....
.....
..... [3]

(ii) Describe how solid sodium conducts electricity.

..... [1]

(b) Some properties of sodium chloride are shown:

- melting point of 801°C
- non-conductor of electricity when solid
- conductor of electricity when molten
- soluble in water.

(i) Name the type of bonding in sodium chloride.

..... [1]

(ii) Explain why sodium chloride conducts electricity when molten.

.....
..... [1]

- (c) A student determines the concentration of a solution of dilute sulfuric acid, H_2SO_4 , by titration with aqueous sodium hydroxide, NaOH .

step 1 25.0 cm³ of 0.200 mol/dm³ NaOH is transferred into a conical flask.

step 2 Three drops of methyl orange indicator are added to the conical flask.

step 3 A burette is filled with H_2SO_4 .

step 4 The acid in the burette is added to the conical flask until the indicator changes colour. The volume of acid is recorded. This process is known as titration.

step 5 The titration is repeated several times until a suitable number of results is obtained.

- (i) Name the piece of apparatus used to measure exactly 25.0 cm³ of 0.200 mol/dm³ NaOH in **step 1**.

..... [1]

- (ii) State the colour change of the methyl orange indicator in **step 4**.

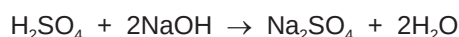
from to [1]

- (iii) State how the student decides that a suitable number of results have been obtained.

.....
..... [1]

- (iv) 20.0 cm³ of H_2SO_4 reacts with 25.0 cm³ of 0.200 mol/dm³ NaOH .

The equation for the reaction is shown.



Calculate the concentration of H_2SO_4 using the following steps.

- Calculate the number of moles in 25.0 cm³ of 0.200 mol/dm³ NaOH .

..... mol

- Determine the number of moles of H_2SO_4 that react with the NaOH .

..... mol

- Calculate the concentration of H_2SO_4 .

..... mol/dm³
[3]

[Total: 12]

2 Calcium hydroxide, $\text{Ca}(\text{OH})_2$, is slightly soluble in water.

(a) Calcium hydroxide can be made by the reaction of calcium with water.

(i) Write the chemical equation for this reaction.

..... [2]

(ii) Name another substance that reacts with water to form calcium hydroxide.

..... [1]

(b) When calcium hydroxide dissolves in water, it dissociates into ions and forms a weakly alkaline solution.

(i) Suggest the pH of aqueous calcium hydroxide.

..... [1]

(ii) Give the formula of the ion responsible for making the solution alkaline.

..... [1]

(c) Limewater is a saturated solution of calcium hydroxide, $\text{Ca}(\text{OH})_2(\text{aq})$.

(i) Name the gas limewater is used to test for.

..... [1]

(ii) Suggest what is meant by the term *saturated solution*.

.....
..... [2]

(iii) Describe how you would make a sample of limewater starting with solid calcium hydroxide.

.....
.....
..... [2]

(iv) Describe how you would test for the presence of calcium ions in a sample of limewater.

test

observations

..... [3]

(d) A 25.0 cm³ sample of limewater is placed in a conical flask. The concentration of Ca(OH)₂ in the limewater is determined by titration with dilute hydrochloric acid, HCl.

(i) Name the item of apparatus used to measure the volume of acid in this titration.

..... [1]

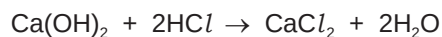
(ii) State the type of reaction which takes place.

..... [1]

(iii) As well as limewater and dilute hydrochloric acid, state what other type of substance must be added to the conical flask.

..... [1]

(iv) The equation for the reaction is shown.



20.0 cm³ of 0.0500 mol/dm³ HCl reacts with the 25.0 cm³ of Ca(OH)₂.

Determine the concentration of Ca(OH)₂ in g/dm³. Use the following steps.

- Calculate the number of moles in 20.0 cm³ of 0.0500 mol/dm³ HCl.

..... mol

- Determine the number of moles of Ca(OH)₂ in 25.0 cm³ of the limewater.

..... mol

- Calculate the concentration of Ca(OH)₂ in mol/dm³.

..... mol/dm³

- Determine the concentration of Ca(OH)₂ in g/dm³.

..... g/dm³
[5]

[Total: 21]

4 A student prepares calcium nitrate, $\text{Ca}(\text{NO}_3)_2$, by adding calcium carbonate to dilute nitric acid.

(a) Write the chemical equation for this reaction.

..... [2]

(b) Describe **two** observations during this reaction.

1

2 [2]

(c) The student continues to add calcium carbonate until it is in excess. The student then removes the excess calcium carbonate by filtration and collects the aqueous calcium nitrate.

State the general term given to a solution collected from filtration.

..... [1]

(d) The student gently heats the aqueous calcium nitrate until the solution is saturated.

(i) Suggest what is meant by the term *saturated solution*.

..... [2]

(ii) Describe how crystals are produced from a hot saturated solution.

..... [1]

- (e) Calcium nitrate crystals are hydrated and have the formula $\text{Ca}(\text{NO}_3)_2 \cdot x\text{H}_2\text{O}$ where x is a whole number of molecules of water.

The student heats the crystals to remove the molecules of water.



- (i) State the term used to describe the calcium nitrate after the molecules of water have been removed.

..... [1]

- (ii) The student heats a sample of $\text{Ca}(\text{NO}_3)_2 \cdot x\text{H}_2\text{O}$ and forms 2.46 g of $\text{Ca}(\text{NO}_3)_2$ and 0.0600 moles of H_2O .

Determine the value of x . Use the following steps.

- Calculate the M_r of $\text{Ca}(\text{NO}_3)_2$.

$M_r =$

- Determine the number of moles of $\text{Ca}(\text{NO}_3)_2$ formed.

moles of $\text{Ca}(\text{NO}_3)_2$ formed =

- Determine the value of x in $\text{Ca}(\text{NO}_3)_2 \cdot x\text{H}_2\text{O}$.

$x =$ [3]

- (f) Nitrates decompose on heating.

Write the chemical equation for the reaction when solid sodium nitrate is heated.

..... [2]

[Total: 14]

2 Sodium is a reactive metal.

(a) Suggest why sodium is stored under oil.

..... [1]

(b) Sodium burns in air to form sodium oxide, Na_2O .

(i) State the term given to a reaction in which a substance burns.

..... [1]

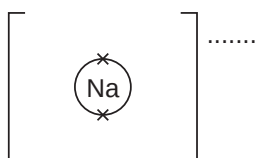
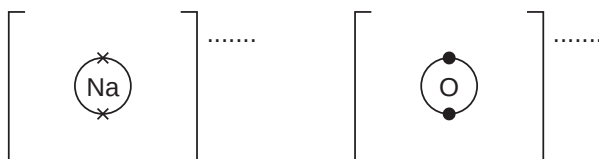
(ii) State the colour of the flame seen when sodium burns.

..... [1]

(iii) Write a chemical equation for the reaction which takes place when sodium burns in air to form sodium oxide.

..... [2]

(iv) Complete the dot-and-cross diagram to show the electron arrangement and charges of the ions in sodium oxide.



[3]

(c) Sodium reacts vigorously with water to form aqueous sodium hydroxide, NaOH, which is a strong base.

(i) Explain in terms of proton transfer what is meant by a base.

..... [1]

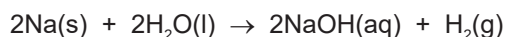
(ii) State a pH number that indicates the presence of a strong alkali.

..... [1]

(iii) State the colour of methyl orange in aqueous sodium hydroxide.

..... [1]

(iv) The equation for the reaction is shown.



Calculate the concentration of NaOH(aq) formed, in g/dm³, when 0.345 g of sodium is added to 50.0 cm³ of distilled water. Assume there is no change in volume.

Use the following steps.

- Calculate the number of moles of Na added.

= mol

- Determine the number of moles of NaOH formed.

= mol

- Calculate the concentration of NaOH in mol/dm³.

concentration of NaOH = mol/dm³

- Determine the M_r of NaOH and calculate the concentration of NaOH in g/dm³.

concentration of NaOH = g/dm³
[5]

(d) When NaOH(aq) is added to aqueous iron(III) chloride, $\text{FeCl}_3\text{(aq)}$, a solid product is formed.

(i) Name the type of reaction where a solid is formed from two solutions.

..... [1]

(ii) State the colour of this solid product.

..... [1]

(iii) Name this solid product.

..... [1]

(iv) Write the ionic equation for the reaction. Include state symbols.

..... [3]

[Total: 22]

- 4 A student prepares magnesium sulfate crystals, MgSO_4 , by adding excess magnesium to dilute sulfuric acid.

(a) Write the chemical equation for this reaction.

..... [1]

(b) Describe **two** observations which show the reaction has finished.

1

2

[2]

(c) The excess magnesium is removed by filtration.

State the general name given to a solid separated from a solution by filtration.

..... [1]

(d) The aqueous magnesium sulfate is heated until crystals begin to appear.

(i) Suggest the name for a solution in which no more solute can dissolve.

..... [1]

(ii) Suggest why more crystals of magnesium sulfate appear on cooling.

..... [1]

(e) Magnesium sulfate crystals have the formula, $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$, where x is a whole number of molecules of water.

The student heats the crystals to remove the molecules of water.



(i) Name the term given to crystals containing molecules of water.

..... [1]

- (ii) The student heats a sample of $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$ and finds it has lost 0.140 moles of H_2O and has 2.40 g of MgSO_4 remaining.

Determine the value of x . Use the following steps.

- Calculate the M_r of MgSO_4 .

$M_r = \dots\dots\dots$

- Determine the number of moles of MgSO_4 formed.

moles of MgSO_4 formed = $\dots\dots\dots$

- Determine the value of x in $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$.

$x = \dots\dots\dots$
[3]

- (f) If the student uses dilute nitric acid instead of dilute sulfuric acid, the salt formed is magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$.

Write the chemical equation for the reaction when solid magnesium nitrate is heated.

$\dots\dots\dots$ [2]

[Total: 12]

5 This question is about sulfuric acid, H_2SO_4 , and salts that can be made from sulfuric acid.

(a) Sulfuric acid is manufactured by the Contact process.

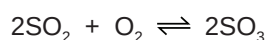
stage 1 Molten sulfur burns in air to produce sulfur dioxide.

stage 2 Sulfur dioxide reacts with oxygen to form sulfur trioxide, SO_3 .

stage 3 Sulfur trioxide reacts with concentrated sulfuric acid to form oleum, $\text{H}_2\text{S}_2\text{O}_7$.

stage 4 Oleum is converted into sulfuric acid.

(i) The equation for the reaction in **stage 2** is shown.



State the temperature and pressure used in **stage 2**.

Name the catalyst used in **stage 2**.

temperature °C

pressure atm

catalyst [3]

(ii) Write the chemical equation for the reaction in **stage 3**.

..... [1]

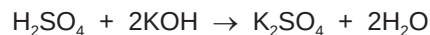
(iii) Name the substance that reacts with oleum in **stage 4**.

..... [1]

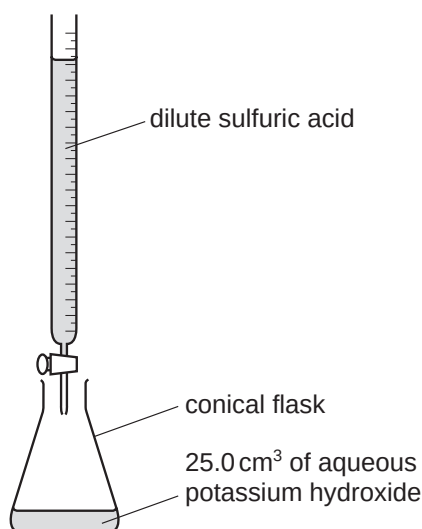
(b) Name the black solid that is produced when concentrated sulfuric acid is added to sugar, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$.

..... [1]

- (c) Dilute sulfuric acid and aqueous potassium hydroxide are used to make aqueous potassium sulfate.



The method includes use of the following apparatus.



- (i) Calculate the volume of 0.0625 mol/dm³ dilute sulfuric acid, H_2SO_4 , that completely reacts with 25.0 cm³ of 0.100 mol/dm³ potassium hydroxide, KOH , to produce aqueous potassium sulfate.

Use the following steps.

- Calculate the number of moles of KOH in 25.0 cm³ of 0.100 mol/dm³ KOH .

= mol

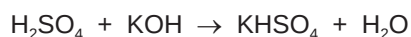
- Deduce the number of moles of H_2SO_4 that react with KOH .

= mol

- Calculate the volume of H_2SO_4 required.

volume = cm^3
[3]

- (ii) The experiment is repeated using the same volume and concentration of potassium hydroxide and the same concentration of dilute sulfuric acid. In this second experiment, the product is aqueous potassium hydrogensulfate, KHSO_4 .



Use your answer to (c)(i) and the equation to deduce the volume of H_2SO_4 required.

volume = cm^3 [1]

- (d) Aqueous potassium hydrogensulfate, $\text{KHSO}_4(\text{aq})$, contains the ions $\text{K}^+(\text{aq})$, $\text{H}^+(\text{aq})$ and $\text{SO}_4^{2-}(\text{aq})$.

Describe the observations in the following tests.

- (i) A flame test is carried out on aqueous potassium hydrogensulfate.

..... [1]

- (ii) Solid copper(II) carbonate is added to aqueous potassium hydrogensulfate.

.....
..... [2]

- (iii) An acidic solution containing aqueous barium ions, $\text{Ba}^{2+}(\text{aq})$, is added to aqueous potassium hydrogensulfate.

..... [1]

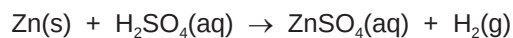
- (e) Write the ionic equation for the reaction in (d)(iii).

Include state symbols.

..... [3]

[Total: 17]

- 4 Hydrogen is produced by the reaction between zinc and dilute sulfuric acid, H_2SO_4 .



- (a) A student carries out an experiment using excess zinc and dilute sulfuric acid.

The student measures the volume of hydrogen produced at regular time intervals using the apparatus shown in Fig. 4.1.

Lumps of zinc are used.

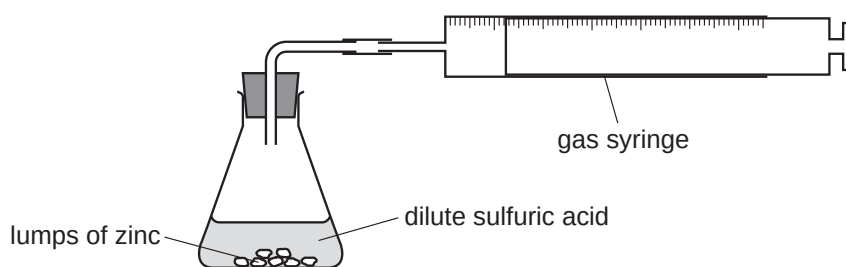


Fig. 4.1

The rate of reaction decreases as the reaction progresses. The rate eventually becomes zero.

- (i) Explain why the rate of reaction decreases as the reaction progresses.

.....
..... [1]

- (ii) Explain why the rate of reaction eventually becomes zero.

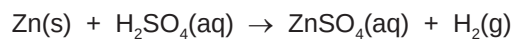
.....
..... [1]

- (b) The experiment is repeated using powdered zinc instead of lumps of zinc.
All other conditions remain the same.

Explain, in terms of collision theory, why the rate of reaction increases if powdered zinc is used.

.....
.....
..... [2]

(c) The equation for the reaction is shown.



25.0 cm³ of 2.00 mol/dm³ H₂SO₄(aq) is added to excess zinc.

Calculate the volume of H₂ formed at room temperature and pressure (r.t.p.). The volume of one mole of any gas is 24 dm³ at r.t.p.

Use the following steps.

- Calculate the number of moles of H₂SO₄ used.

..... mol

- Deduce the number of moles of H₂ produced.

..... mol

- Calculate the volume of H₂ formed at r.t.p.

..... dm³
[3]

(d) Hydrogen can also be produced by the reaction of zinc with dilute hydrochloric acid.

- (i) Write a symbol equation for this reaction.

..... [2]

- (ii) State the test for hydrogen gas.

test

positive result
[1]

[Total: 10]

- 4 Aqueous hydrogen peroxide, H_2O_2 , slowly forms water and oxygen at room temperature and pressure, r.t.p. This reaction is catalysed by manganese(IV) oxide.

The equation is shown.



- (a) State the test for oxygen gas.

test

observations

[1]

- (b) A student investigates the rate of formation of oxygen gas when manganese(IV) oxide is added to aqueous hydrogen peroxide.

The volume of oxygen gas formed is measured at regular time intervals at r.t.p. The results are plotted onto the graph in Fig. 4.1.

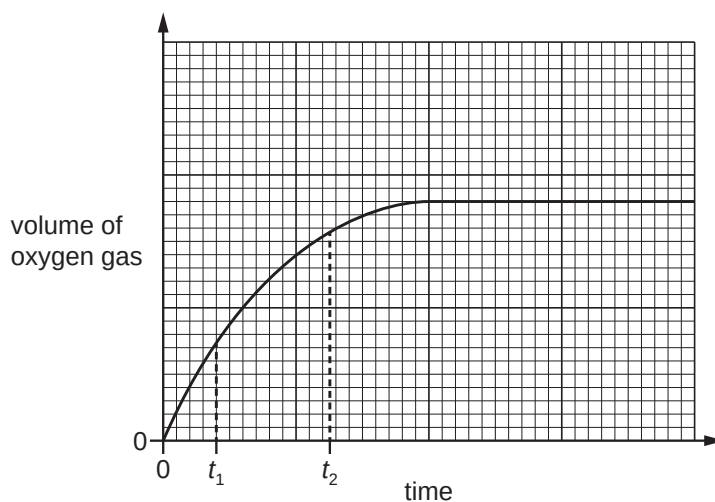


Fig. 4.1

- (i) State how the graph in Fig. 4.1 shows the rate of reaction at time t_2 , is lower than at time t_1 .

[1]

- (ii) Explain, using collision theory, why the rate of reaction at time t_2 is lower than at time t_1 .

[2]

- (iii) On Fig. 4.1, sketch the graph obtained when the experiment is repeated using aqueous hydrogen peroxide at a higher temperature. All other conditions remain the same.

[2]

- (c) Manganese(IV) oxide is added to 20 cm³ of aqueous hydrogen peroxide. The total volume of oxygen gas produced is 72 cm³ at r.t.p.



Calculate the concentration of the aqueous hydrogen peroxide in g/dm³ using the following steps.

- Calculate the number of moles of oxygen gas produced.

..... mol

- Determine the number of moles of hydrogen peroxide which reacts.

..... mol

- Calculate the concentration of aqueous hydrogen peroxide in mol/dm³.

..... mol/dm³

- Calculate the concentration of aqueous hydrogen peroxide in g/dm³.

..... g/dm³
[5]

- (d) Suggest the identity of one **other** metal oxide which also catalyses this reaction.

..... [1]

[Total: 12]

5 (a) Lead(II) bromide, PbBr_2 , is an insoluble salt and is made by precipitation.

(i) Name **two** aqueous solutions that produce a precipitate of lead(II) bromide when they are mixed.

1

2 [2]

(ii) Describe how to produce a pure sample of lead(II) bromide from the mixture of aqueous solutions in (a)(i).

.....

..... [2]

(iii) Write an ionic equation for the precipitation reaction which produces lead(II) bromide. Include state symbols.

..... [3]

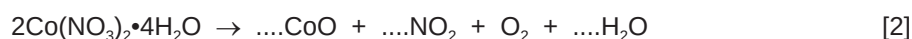
(b) When iron(II) sulfate crystals are heated strongly, sulfur dioxide gas is given off.

Describe a test for sulfur dioxide gas.

test

observations [2]

(c) Complete the equation for the thermal decomposition of hydrated cobalt(II) nitrate.



(d) Hydrated cobalt(II) sulfate, $\text{CoSO}_4 \cdot x\text{H}_2\text{O}$, produces water when it is heated.



A student does an experiment to determine the value of x in $\text{CoSO}_4 \cdot x\text{H}_2\text{O}$.

step 1 The student weighs a sample of hydrated cobalt(II) sulfate.

step 2 The student heats the sample of hydrated cobalt(II) sulfate.

step 3 The student weighs the remaining solid after heating.

(i) Describe what else the student should do to ensure that **all** the water has been given off. No other substances are required.

.....
.....
..... [2]

(ii) In an experiment, 1.405 g of $\text{CoSO}_4 \cdot x\text{H}_2\text{O}$ is heated until all the water is given off.

The mass of CoSO_4 that remains is 0.775 g.

[M_r : CoSO_4 , 155; H_2O , 18]

Determine the value of x using the following steps.

- Calculate the number of moles of CoSO_4 that remains.

..... mol

- Calculate the mass of H_2O given off.

..... g

- Calculate the number of moles of H_2O given off.

..... mol

- Determine the value of x .

$x =$
[4]

[Total: 17]

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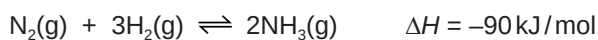
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- 3 The symbol equation for the industrial production of ammonia is shown.



- (a) Name this industrial process.

..... [1]

- (b) State the meaning of ΔH .

..... [1]

- (c) State the typical conditions and name the catalyst used in the industrial production of ammonia.

temperature and units

pressure and units

catalyst used

[3]

- (d) State **two** methods of increasing the rate of this reaction.

1

2

[2]



b

(e) The symbol equation for the reaction can be represented as shown in Fig. 3.1.

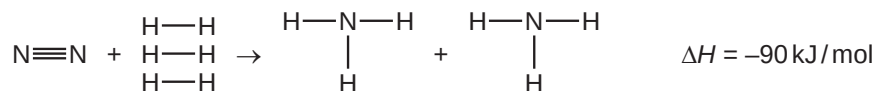


Fig. 3.1

Table 3.1 shows some bond energies.

Table 3.1

bond	N≡N	H–H
bond energy in kJ/mol	945	435

Use the bond energies in Table 3.1 and ΔH to calculate the bond energy of an N–H bond, in kJ/mol.

Use the following steps.

- Calculate the energy needed to break bonds in the reactants.

..... kJ

- Calculate the energy released when bonds form in the products.

..... kJ

- Calculate the energy of an N–H bond.

..... kJ/mol
[3]

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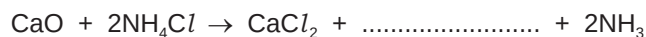
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(f) An incomplete symbol equation for the preparation of ammonia in the laboratory is shown.



(i) Complete the symbol equation. [1]

(ii) Name NH_4Cl .

..... [1]

(iii) Calculate the volume of ammonia, NH_3 , measured at room temperature and pressure, which forms when 1.12 g of CaO is heated with excess NH_4Cl .
[M_r : CaO , 56]

..... cm^3 [3]

[Total: 15]

4 When magnesium nitrate is heated strongly, magnesium oxide is formed.

(a) The equation for this reaction is shown.



(i) State the change in oxidation number of nitrogen, N, in this reaction.

from to [2]

(ii) Identify the element which is oxidised in this reaction.

..... [1]

(iii) Calculate the volume of NO_2 gas, at r.t.p., formed when 7.40 g of $\text{Mg}(\text{NO}_3)_2$ is heated.

Use the following steps.

- Calculate the M_r of $\text{Mg}(\text{NO}_3)_2$.

.....

- Calculate the number of moles of $\text{Mg}(\text{NO}_3)_2$ used.

..... mol

- Determine the number of moles of NO_2 formed.

..... mol

- Calculate the volume of NO_2 gas, in cm^3 , at r.t.p.

..... cm^3
[4]

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(b) Magnesium oxide, MgO , is an ionic compound.

Complete the dot-and-cross diagram in Fig. 4.1 of the ions in magnesium oxide.

Give the charges on each of the ions.

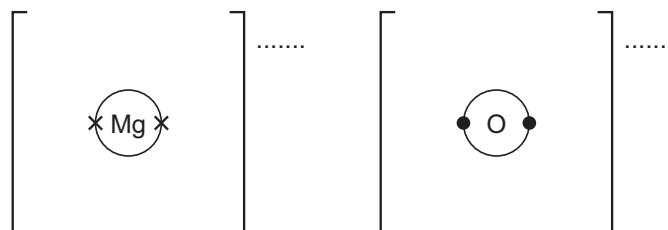


Fig. 4.1

[3]

(c) Oxygen is a covalent molecule.

Complete the dot-and-cross diagram in Fig. 4.2 of a molecule of oxygen.
The inner shells have been drawn.

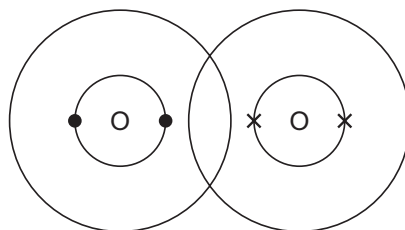


Fig. 4.2

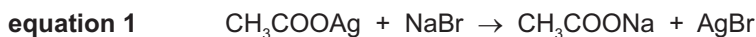
[2]

[Total: 12]



- 4 Silver bromide, AgBr, is made when aqueous silver ethanoate, CH₃COOAg, is added to aqueous sodium bromide, NaBr.

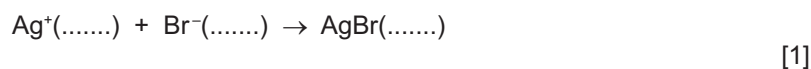
The equation for the reaction is shown in **equation 1**.



The method includes the following steps.

- step 1** Add 200.0 cm³ of 0.0500 mol/dm³ CH₃COOAg to a beaker.
This volume contains 0.0100 mol of Ag⁺ ions.
- step 2** Add 50.0 cm³ of aqueous NaBr. This volume contains 0.0100 mol of Br⁻ ions.
A precipitate forms.
- step 3** Filter the mixture.
- step 4** Dry the solid residue until all the water is removed.
- step 5** Record the mass of the dry residue.

- (a) Complete the ionic equation for the reaction by adding the missing state symbols.



- (b) Name a different aqueous silver salt which could be used in **step 1**.

..... [1]

- (c) Use the information in **step 2** to calculate the concentration of aqueous NaBr.

concentration = mol/dm³ [1]

- (d) State the colour of the precipitate which forms in **step 2**.

..... [1]



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- (e) Use the information in **step 1**, **step 2** and **equation 1** to determine the number of moles of AgBr formed. Use this value to calculate the mass of AgBr formed.

number of moles of AgBr =

mass of AgBr = g
[3]

- (f) Name the salt dissolved in the filtrate in **step 3**.

..... [1]

- (g) The recorded mass of the dry residue in **step 5** is greater than the mass calculated in (e) because a step is missing from the procedure.

- (i) Suggest the missing step.

..... [1]

- (ii) Name the substance responsible for the greater mass of the dry residue.

..... [1]

- (h) Barium sulfate can be made by the same method but with different aqueous solutions.

- (i) Suggest **two** aqueous solutions which can be added together to make barium sulfate.

..... and [2]

- (ii) Write the balanced symbol equation for this reaction.

..... [2]

[Total: 14]



4 This question is about nitrogen and its compounds.

Nitrogen contains molecules with the formula N_2 .

- (a) Complete the dot-and-cross diagram in Fig. 4.1 to show the electronic configuration in a nitrogen molecule. Show outer-shell electrons only.

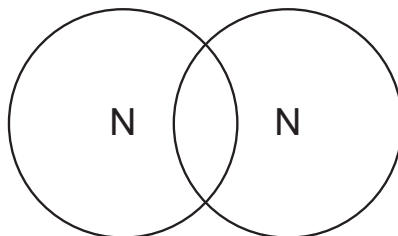


Fig. 4.1

[2]

- (b) Ammonia, NH_3 , is manufactured by reacting nitrogen with hydrogen, H_2 , in the Haber process.

- (i) State **three** typical conditions for the reaction between nitrogen and hydrogen in the Haber process.

- 1
- 2
- 3

[3]

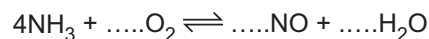
- (ii) Write the symbol equation for the chemical reaction in the Haber process.

..... [1]

- (c) Ammonia is converted into nitric acid in a two-step process.

In step 1, ammonia and oxygen are passed over a catalyst.

- (i) Balance the symbol equation for this reaction.



[1]

- (ii) Give the oxidation number of nitrogen in:

- NH_3
- NO

[2]



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(iii) Define oxidation in terms of oxidation number.

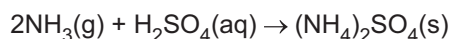
..... [1]

(iv) In step 2, oxygen and water react with NO to produce nitric acid as the only product.

Write a symbol equation for this chemical reaction.

..... [2]

(d) Ammonia is converted into ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$.



360 dm³ of ammonia gas, measured at r.t.p., reacts with excess sulfuric acid.

Calculate the mass of ammonium sulfate produced, using the following steps.

- Calculate the number of moles of $\text{NH}_3(\text{g})$ in 360 dm³.

One mole of any gas occupies 24 dm³ at r.t.p.

..... mol

- Calculate the number of moles of $(\text{NH}_4)_2\text{SO}_4$ produced.

..... mol

- Calculate the mass of $(\text{NH}_4)_2\text{SO}_4$ produced.

The M_r of $(\text{NH}_4)_2\text{SO}_4$ is 132.

..... g
[3]

[Total: 15]

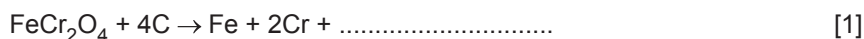


4 Chromium is the element with atomic number 24 in the Periodic Table.

(a) The main ore of chromium is chromite. Chromite contains FeCr_2O_4 .

FeCr_2O_4 reacts with carbon.

(i) Complete the equation for this reaction.



(ii) Suggest **one** disadvantage of extracting chromium by reacting chromite with carbon.

..... [1]

(b) Chromium can be mixed with nickel and other elements to form stainless steel.

Name the type of substance formed when a metal is mixed with other elements.

..... [1]

(c) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ is a compound containing chromium.

The negative ion in $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ is $\text{Cr}_2\text{O}_7^{2-}$.

(i) State the sum of the oxidation numbers in the $\text{Cr}_2\text{O}_7^{2-}$ ion.

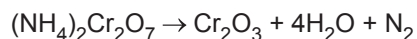
..... [1]

(ii) The oxidation number of each O in $\text{Cr}_2\text{O}_7^{2-}$ ions is -2 .

Determine the oxidation number of each Cr in $\text{Cr}_2\text{O}_7^{2-}$ ions. Show your working.

..... [2]

(iii) When $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ is heated the following reaction occurs.



Calculate the volume of nitrogen gas produced at r.t.p., in cm^3 , when 1.26 g of $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ is heated using the following steps.

The M_r of $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ is 252.

- Calculate the number of moles of $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ used.

..... mol

- Determine the number of moles of N_2 formed.

..... mol

- Calculate the volume of N_2 formed at r.t.p. in cm^3 .

..... cm^3
[3]

[Total: 9]

4 This question is about compounds of sulfur.

(a) Sulfuric acid is manufactured in a four-stage process.

stage 1 Sulfur dioxide is produced from ores containing sulfur.

stage 2 Sulfur dioxide reacts with oxygen to form sulfur trioxide.

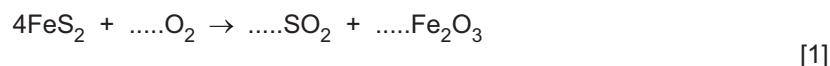
stage 3 Sulfur trioxide reacts with concentrated sulfuric acid to form oleum, $\text{H}_2\text{S}_2\text{O}_7$.

stage 4 Oleum reacts with water to form concentrated sulfuric acid.

(i) Iron pyrite is an ore containing sulfur. The ore contains a compound with the formula FeS_2 .

FeS_2 reacts with oxygen in the air to produce sulfur dioxide.

Balance the symbol equation for this reaction.



(ii) Complete the dot-and-cross diagram in Fig. 4.1 to show the electronic configuration in an oxygen molecule, O_2 . Show outer-shell electrons only.

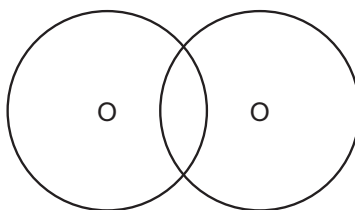


Fig. 4.1

[2]

(b) (i) State **three** typical conditions for the reaction between sulfur dioxide and oxygen in **stage 2**.

1

2

3

[3]



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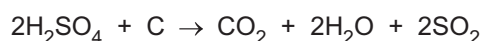
(ii) Write a symbol equation for the reaction that occurs in **stage 2**.

..... [1]

(iii) Write the symbol equation for the reaction in **stage 3**.

..... [1]

(c) Sulfuric acid reacts with carbon. The equation for the reaction is shown.



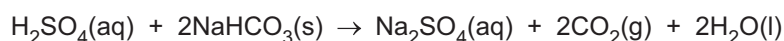
Give the oxidation number of carbon in:

• C

• CO_2

[2]

(d) Dilute sulfuric acid reacts with sodium hydrogencarbonate, NaHCO_3 , to produce carbon dioxide gas, CO_2 .



4.20 g of NaHCO_3 is added to excess dilute sulfuric acid.

Calculate the volume of $\text{CO}_2(\text{g})$, measured at r.t.p., produced using the following steps.

- Calculate the number of moles of NaHCO_3 in 4.20 g.

The M_r of NaHCO_3 is 84.

..... mol

- Deduce the number of moles of $\text{CO}_2(\text{g})$ that are produced.

..... mol

- Calculate the volume of $\text{CO}_2(\text{g})$ produced in cm^3 .

One mole of any gas occupies $24\,000\text{ cm}^3$ at r.t.p.

..... cm^3
[3]

[Total: 13]



2 Calcium is an element in Group II of the Periodic Table.

(a) Identify the element in Group II which has only five occupied electron shells.

..... [1]

(b) Name and describe the type of bonding found in elements in Group II.

name

description

.....

.....

.....

[4]

(c) When a piece of calcium is added to some cold water containing universal indicator a reaction takes place.

(i) Give **three** observations when this reaction takes place.

1

2

3

[3]

(ii) Name the **two** products of this reaction.

..... and [2]

(d) Calcium burns in oxygen.

(i) State the colour of the flame.

..... [1]

(ii) Write the symbol equation for this reaction.

..... [2]

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- (e) Crystals of hydrated calcium nitrate contain water molecules.

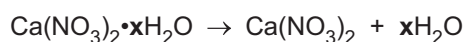
The formula of hydrated calcium nitrate is $\text{Ca}(\text{NO}_3)_2 \cdot x\text{H}_2\text{O}$.

x is a whole number.

- (i) State the term given to the water molecules present in hydrated crystals.

..... [1]

- (ii) When hydrated calcium nitrate is heated gently, the following reaction occurs.



A sample of hydrated calcium nitrate is heated gently. 3.28 g of $\text{Ca}(\text{NO}_3)_2$ forms and the mass of the crystals decreases by 1.44 g.

[M_r : $\text{Ca}(\text{NO}_3)_2$, 164; H_2O , 18]

Determine the value of x in $\text{Ca}(\text{NO}_3)_2 \cdot x\text{H}_2\text{O}$ using the following steps.

- Calculate the number of moles of $\text{Ca}(\text{NO}_3)_2$ that remain.

..... mol

- Calculate the number of moles of H_2O given off.

..... mol

- Determine the value of x .

$x =$ [3]

[Total: 17]



- 3 A student makes crystals of the salt sodium sulfate, Na_2SO_4 . The student reacts 0.200 mol/dm^3 dilute sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$, with aqueous sodium hydroxide, $\text{NaOH}(\text{aq})$.

The student uses the following steps.

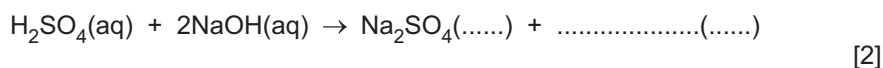
step 1 The student places 40.0 cm^3 of $\text{NaOH}(\text{aq})$ into a conical flask. This volume contains 0.0100 moles of NaOH .

step 2 The student adds a few drops of methyl orange indicator to the $\text{NaOH}(\text{aq})$ in the conical flask.

step 3 The student adds 0.200 mol/dm^3 $\text{H}_2\text{SO}_4(\text{aq})$ to the flask until the end-point is reached.

step 4 The student transfers the mixture from the conical flask to an evaporating basin and obtains dry crystals.

- (a) Complete the symbol equation for the reaction. Include state symbols.



- (b) State the type of exothermic reaction taking place.

..... [1]

- (c) Calculate the concentration of $\text{NaOH}(\text{aq})$ used in **step 1**.

concentration of $\text{NaOH}(\text{aq}) = \text{..... mol/dm}^3$ [1]

- (d) Name the item of apparatus the student uses to add $\text{H}_2\text{SO}_4(\text{aq})$ in **step 3**.

..... [1]

- (e) Calculate the volume of $\text{H}_2\text{SO}_4(\text{aq})$, in cm^3 , added in **step 3**.

volume of $\text{H}_2\text{SO}_4(\text{aq}) = \text{..... cm}^3$ [2]

- (f) State the colour change observed in **step 3**.

from to [2]



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- (g) The dry crystals formed in **step 4** are coloured and **not** white. This is because the student should do an additional step between **step 3** and **step 4**.

Suggest what the student should do in this additional step to produce white crystals.

.....
..... [1]

- (h) In **step 4**, the student gently heats the solution in the evaporating basin until the solution is saturated. The student then stops heating and leaves the hot solution to cool. Crystals start to appear.

- (i) Explain the term saturated solution.

.....
.....
..... [2]

- (ii) Explain why crystals start to appear as the hot solution cools.

.....
..... [1]

- (iii) Suggest the effect, if any, on the mass of crystals collected in **step 4** if the solution in the evaporating basin is allowed to dry without gentle heating.

..... [1]

[Total: 14]



- 3 A student prepares a sample of the insoluble salt barium sulfate, BaSO_4 .
The student adds aqueous sodium sulfate, $\text{Na}_2\text{SO}_4(\text{aq})$, to aqueous barium chloride, $\text{BaCl}_2(\text{aq})$.

The student carries out the following steps.

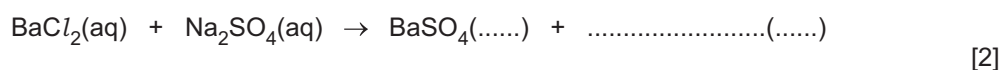
step 1 Dissolve 0.100 moles of solid BaCl_2 in 100cm^3 of distilled water in a beaker to form $\text{BaCl}_2(\text{aq})$.

step 2 Add an aqueous solution which contains 0.100 moles of $\text{Na}_2\text{SO}_4(\text{aq})$ to the beaker containing the $\text{BaCl}_2(\text{aq})$ from **step 1**.

step 3 Stir the mixture and then filter it.

step 4 Dry the solid left in the filter paper.

- (a) Complete the symbol equation for the reaction. Include state symbols.



- (b) Calculate the mass of solid BaCl_2 dissolved in **step 1**.

mass = g [2]

- (c) The concentration of $\text{Na}_2\text{SO}_4(\text{aq})$ in **step 2** is 1.25mol/dm^3 .

Calculate the volume, in cm^3 , of 1.25mol/dm^3 $\text{Na}_2\text{SO}_4(\text{aq})$ added in **step 2** which contains 0.100 moles of Na_2SO_4 .

volume = cm^3 [1]

- (d) State the colour of the solid formed in **step 2**.

..... [1]

- (e) The mixture is filtered in **step 3**.

State the general term given to a solid left in the filter paper after filtration.

..... [1]



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- (f) The actual mass of dry BaSO_4 collected in **step 4** is greater than the expected mass. This is because the student should do an additional step between **step 3** and **step 4**.

Suggest what the student should do in this additional step, and explain why the actual mass collected is greater than the expected mass.

.....
.....
.....
..... [2]

- (g) State the name of this method of salt preparation.

..... [1]

- (h) State the name of a different:

- barium salt that can be used in place of BaCl_2
- sulfate salt that can be used in place of Na_2SO_4
- barium salt that can be made by this method.

[3]

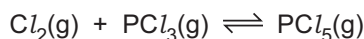
[Total: 13]



4 This question is about chlorine and compounds of chlorine.

- (a) Gaseous chlorine reacts with gaseous phosphorus(III) chloride to form gaseous phosphorus(V) chloride. The reaction is a reversible reaction.

When the three gases are in a closed container the system reaches equilibrium.



- (i) Describe a reversible reaction at equilibrium in terms of:

- the rate of the forward reaction and the rate of the reverse reaction

.....

- the concentrations of reactants and products.

.....

[2]

- (ii) Complete Table 4.1 using only the words **increases**, **decreases** or **no change**.

Table 4.1

change to condition	effect on the rate of the forward reaction	effect on the equilibrium concentration of PCl_5
the pressure is decreased		
a catalyst is added	increases	

[3]

- (iii) When the temperature of the equilibrium mixture is increased, the equilibrium concentration of PCl_5 decreases.

State what conclusion about the forward reaction can be made from this information.

..... [1]





(b) Ethene, C_2H_4 , reacts with chlorine, Cl_2 , to form 1,2-dichloroethane, CH_2ClCH_2Cl .

The equation for this reaction can be represented as shown.

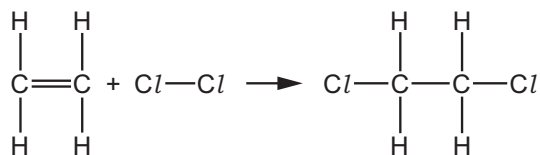


Table 4.2 shows some bond energies.

Table 4.2

bond	bond energy in kJ/mol
C–C	350
C=C	610
C–H	410
Cl–Cl	240
C–Cl	340

Use the bond energies in Table 4.2 to calculate the enthalpy change, in kJ/mol, of the reaction.

Use the following steps.

- Calculate the total energy needed to break the bonds in C_2H_4 and Cl_2 .

..... kJ

- Calculate the total energy released when the bonds form in CH_2ClCH_2Cl .

..... kJ





- Calculate the enthalpy change of the reaction.
Your answer should include a sign.

..... kJ/mol
[3]

- (c) Chlorine reacts with nitrogen to form nitrogen trichloride, NCl_3 .

Complete the dot-and-cross diagram in Fig. 4.1 of a molecule of NCl_3 .

Show outer shell electrons only.

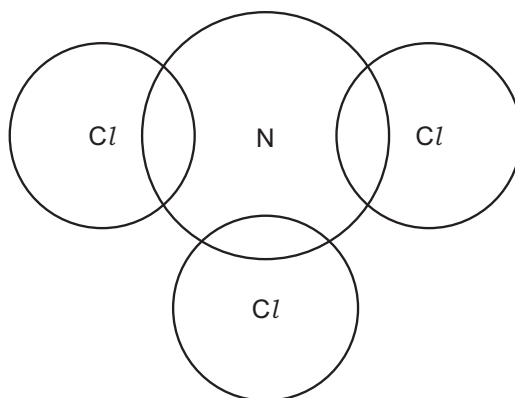


Fig. 4.1

[3]





- (d) When solid magnesium carbonate, MgCO_3 , is added to dilute hydrochloric acid, HCl , a chemical reaction occurs. The equation for the reaction is shown.



- (i) Give **two** observations when solid magnesium carbonate is added to dilute hydrochloric acid.

1

2 [2]

- (ii) Calculate the volume, in cm^3 , of $\text{CO}_2(\text{g})$ that is produced at room temperature and pressure (r.t.p.) when 50.0 cm^3 of $0.100 \text{ mol/dm}^3 \text{ HCl}$ reacts with excess MgCO_3 .

The volume of 1 mol of any gas is $24\,000 \text{ cm}^3$ at r.t.p.

Use the following steps.

- Calculate the number of moles of HCl in 50.0 cm^3 of $0.100 \text{ mol/dm}^3 \text{ HCl}$.

..... mol

- Deduce the number of moles of $\text{CO}_2(\text{g})$ produced.

..... mol

- Calculate the volume, in cm^3 , of $\text{CO}_2(\text{g})$ produced at r.t.p.

..... cm^3
[3]

[Total: 17]

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